Sinie van der Ben

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Research Profile

PhD Researcher at ETH Zurich, Underlying all my research is an interest driven by how models organise and encode knowledge, concepts and representations, originating from a fascination for neurosciences that try to map structures to the brain. With the outcome that if internal mechanisms can be understood, audited and trusted, an attempt can be made to make AI safe.

Education

PhD Artificial Intelligence — Model Interpretability Nov 2024 – Present
ETH Zurich, IVIA Lab
Zurich, Switzerland

- Investigated limitations of SAE evaluation methods: auto-interpretability is brittle under distribution shift
- Currently looking into the potential influence of spectral position on a priori steerability information
- Generally interested in *how models store knowledge and how we can retrieve this.*
- Supervised by Prof. Mennatallah El-Assady (ETH AI Center)

MSc Artificial Intelligence — With Distinction (GPA 8.6/10) Sep 2020 – Jul 2023
Utrecht University · Thesis conducted at ETH Zurich

- Thesis: “Explainability for Network Anomaly Detection” (grade 8.7/10) — quantitative XAI evaluation for anomaly detection systems
- Descartes Honours College — selective programme for top students across disciplines

BSc Artificial Intelligence (GPA 8.0/10) Sep 2016 – Apr 2020
Utrecht University

Publications & Research Output

- van der Ben, S., Roch, N., Hedström, A., & El-Assady, M. (2026). — Building fast, evaluating slow: Pipeline choices dominate autointerpretability score variance. *Accepted as poster at ICML 2026 MechInterpret Workshop*
- van der Ben, S., Baur, R., Metz, Y., & El-Assady, M. (2026). Where Do Models Find Happiness? Emotion Vectors in Open-source LLMs. *Accepted as virtual poster at ICML 2026 MechInterpret Workshop*
- Yan X., Sevastjanova R., van der Ben S., El-Assady M., Wang B. — Explainable Mapper: Charting LLM Embedding Spaces Using Perturbation-Based Explanation and Verification Agents. arXiv preprint, 2025

Technical Contributions

- Contributed to TransformerLens by integrating support for [Apertus] (Swiss AI LLM). [Link] to Pull Request.

Research & Industry Experience

Research Assistant Oct 2023 – Nov 2024
Utrecht University, Utrecht, the Netherlands

- Applying ML methods to study dietary patterns and ultra-processed food classification
- Cross-institutional collaboration between Utrecht University, the Dutch National Institute for Public Health (RIVM), and Harvard Medical School

Machine Learning Engineer Nov 2023 – Sep 2024
Enjins, Utrecht, the Netherlands

- Designed and deployed ML pipelines and data platform infrastructure (Python, Terraform, Kubernetes, AWS EKS, Azure)
- Contributed to data integration proposals and cloud architecture; built Azure data platform from the ground up

Deep Learning Engineering Intern - Multimodal Deepfake Detection Feb 2022 – Aug 2022
DuckDuckGoose AI, Delft, the Netherlands

- Implemented a dual-stream multimodal architecture for deepfake detection, with separate vision and language encoders connected by intermediate cross-modal attention layers

Working Student — Risk Advisory & Cloud Engineering Nov 2022 – Oct 2023
Deloitte, Amsterdam, the Netherlands

- Supported creation of a DevSecOps module for the Risk Advisory team
- Contributed to a workshop series on Serverless Cloud Computing (Terraform, Golang, AWS)

Deep Learning Intern — Cancer Cell Detection Feb 2021 – Jul 2021
Prinses Máxima Centrum, Utrecht, the Netherlands

- Applied GANs to augment training data for fluorescent marker-limited cancer cell detection, improving classifier performance under low-data regimes

Student Assistant — PROVEE Project Jan 2021 – Dec 2021
Utrecht University

- Contributed to PROVEE (Progressive Explainable Embeddings), developing incremental PCA-based visualisation of high-dimensional ML embeddings

Hackathon

Started with hackathons in 2026! Planning to do more.

ETH AI Centre x Anthropic Sprint June 2026

- Started from scratch to inmplement a real-time tool that listens to a group conversation and lets a data chart **emerge from the discussion**. It transcribes speaker-attributed speech, watches the conversation, and continuously (re)generates the chart as model-written Python rendered via Anthropic's Code Execution sandbox.

Future of Health Hackathon May 2026

- Improve the workflow of a pathologist retrieving loads of information about tests results of patients with various testing profiles. Implemented an interactive graph to provide an instant visual overview

Technical Skills

Core: Python, PyTorch, C++ (basic), Golang (basic)
ML: Deep learning, multimodality, XAI methods, SAEs
Infrastructure: Terraform, Kubernetes, AWS EKS, Azure, Golang, Docker
Languages: Dutch (native), English (fluent), German (intermediate)

Other

Competitive Rower Nov 2018 – Jun 2023

- Trained 7–10 sessions per week at national competitive level; elected Secretary of Career Days, organising a week of speaker events sponsored by Philips